

Kerberos Authentication



Kerberos

- The name Kerberos was taken from Greek mythology; it was a three-headed dog who guarded the gates of hades.
- The name Kerberos we are going to talk about is an authentication protocol used to verify the identity of a user or host in a network.



Kerberos

- Kerberos was originally developed at MIT.
- Most modern operating systems such as Windows, Linux, and Macintosh recently have the Kerberos protocol implemented.
- It is the default authentication protocol for network logon in Microsoft Windows.

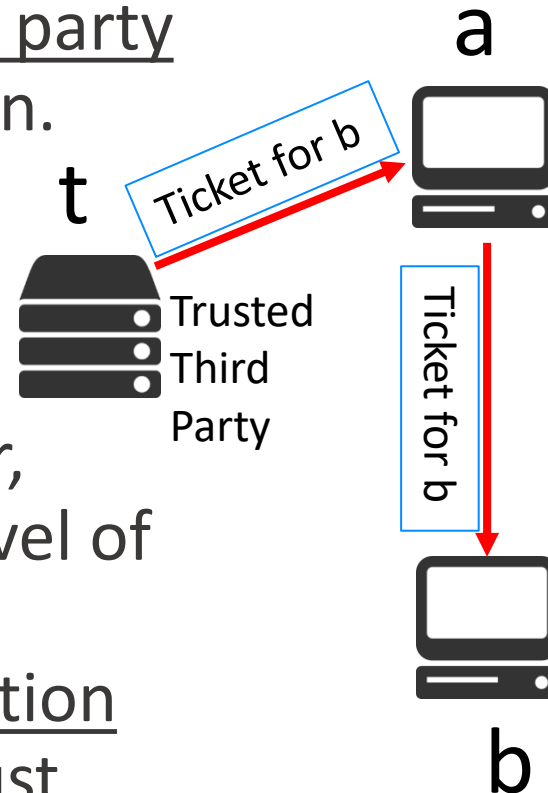
Symmetric Encryption

- Kerberos uses symmetric encryption method.
- Symmetric encryption is a data encryption method whereby the same key is used to encode and decode information.

Trusted Third Party

- Kerberos uses a trusted third party as the basis for authentication.

- A trusted third party is an independent service provider, assumed to have a certain level of trust. The trusted third party facilitates secure communication between two parties who trust this third party.

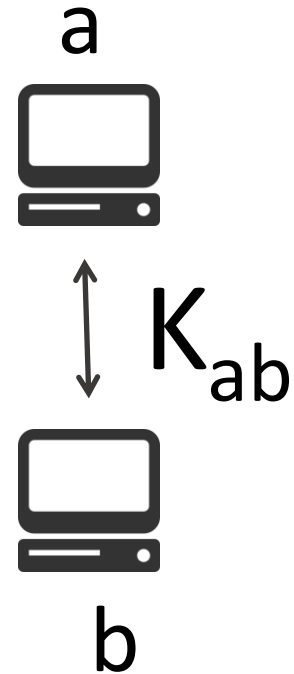


Authentication

- In Kerberos, the authentication between a pair of nodes is done by using a key.

{ message }_{K_{ab}}

- A symmetric key provides mutual authentication between a pair of nodes.

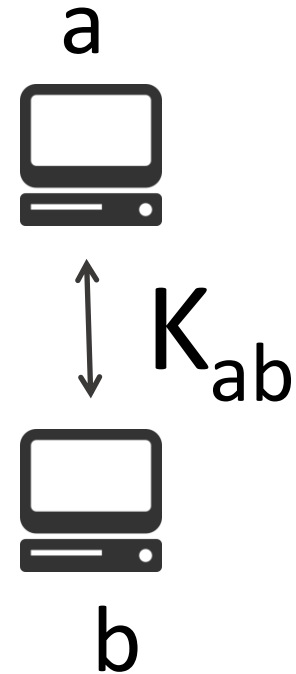


Authentication

- The key must be kept secret from other nodes, hence the term secret key.

{ message } K_{ab}

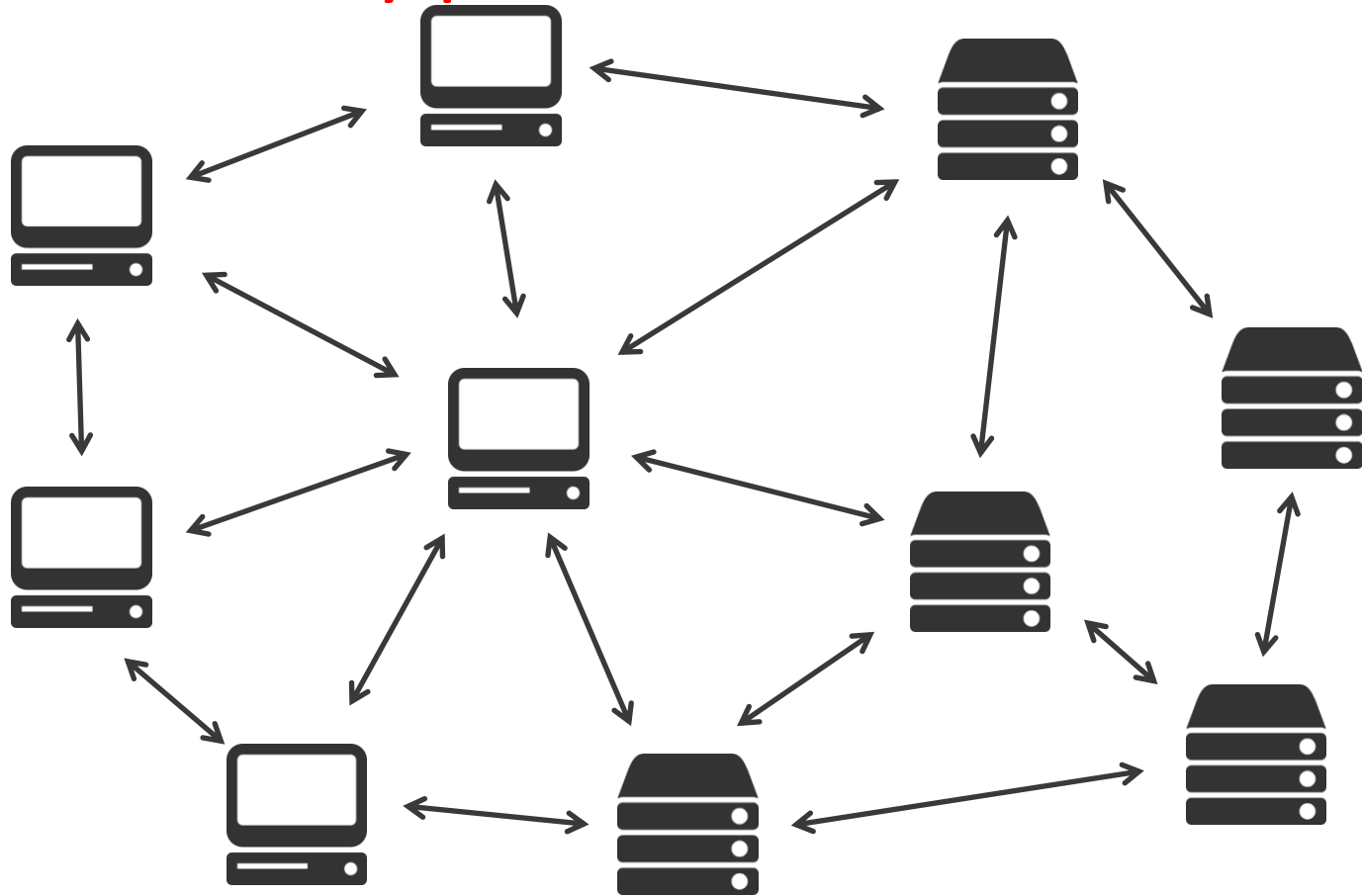
- Password is the most common form of secret key.



What happened when we have a
large enterprise-wide local area
network?

Encrypting Large-scale Networks

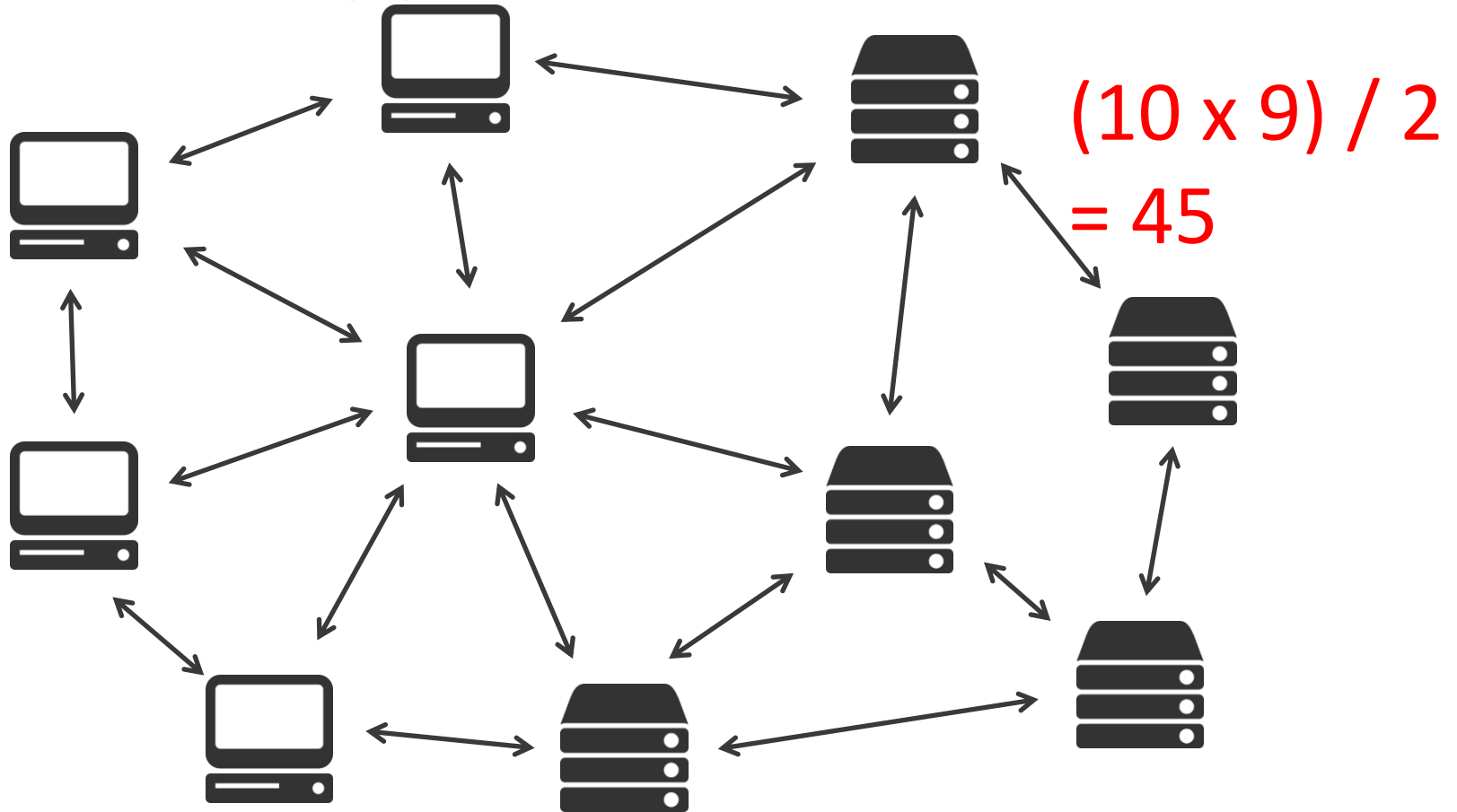
How many pairs of nodes are there?



How many secret keys are needed?

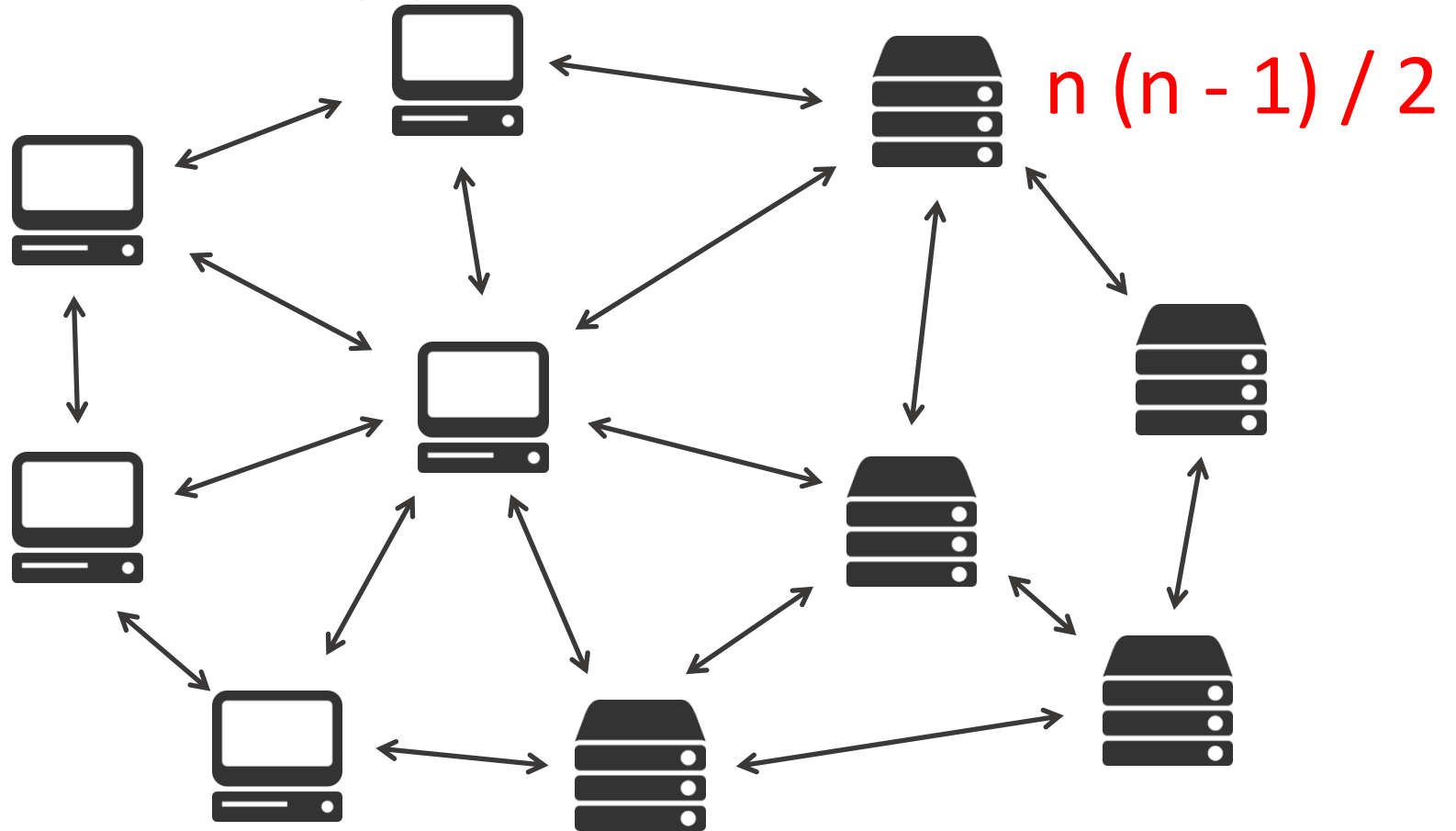
Encrypting Large-scale Networks

How many pairs of nodes are there?



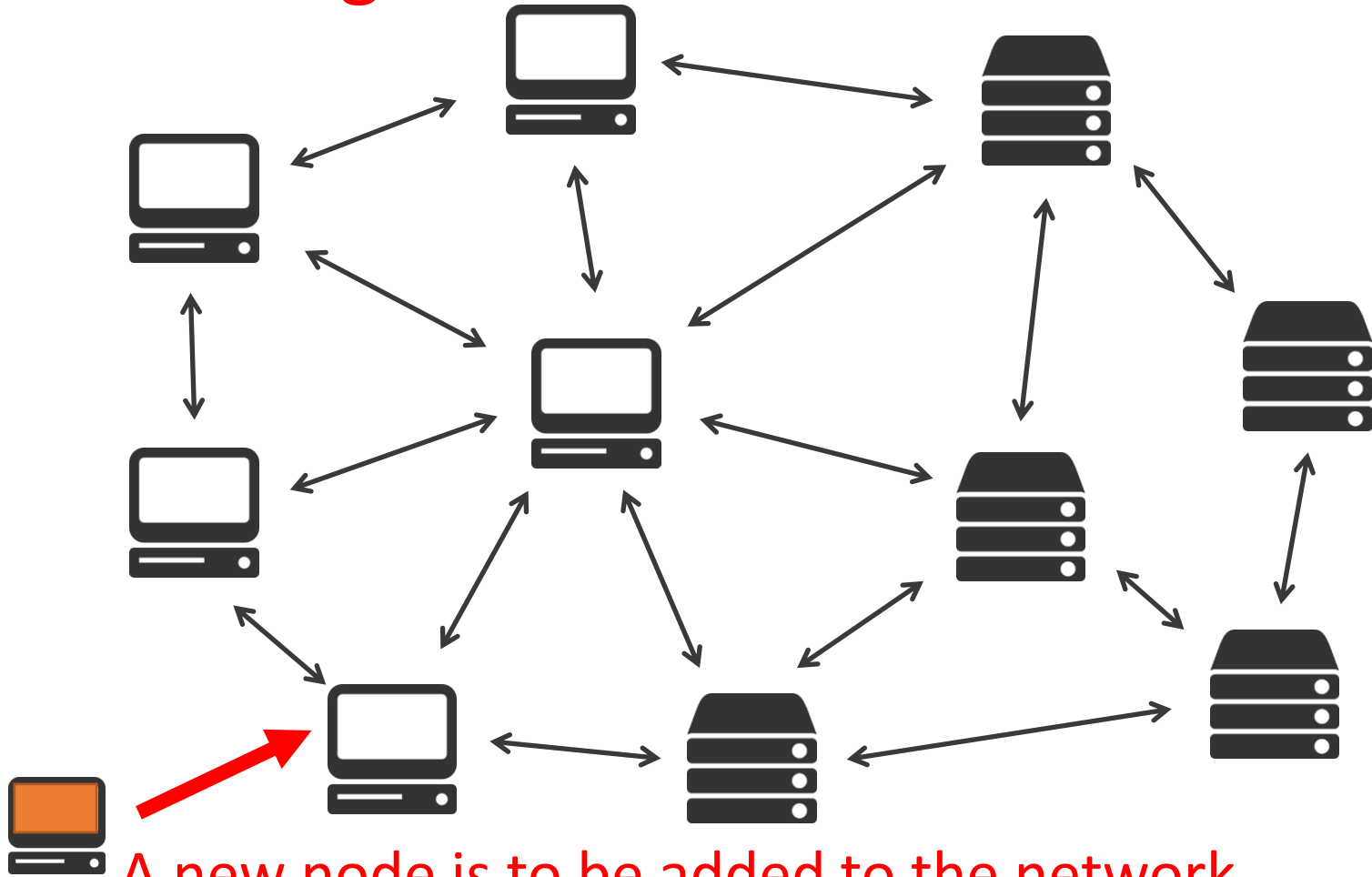
Encrypting Large-scale Networks

How many pairs of nodes are there?



Encrypting Large-scale Networks

Imagine there are 1000 nodes.

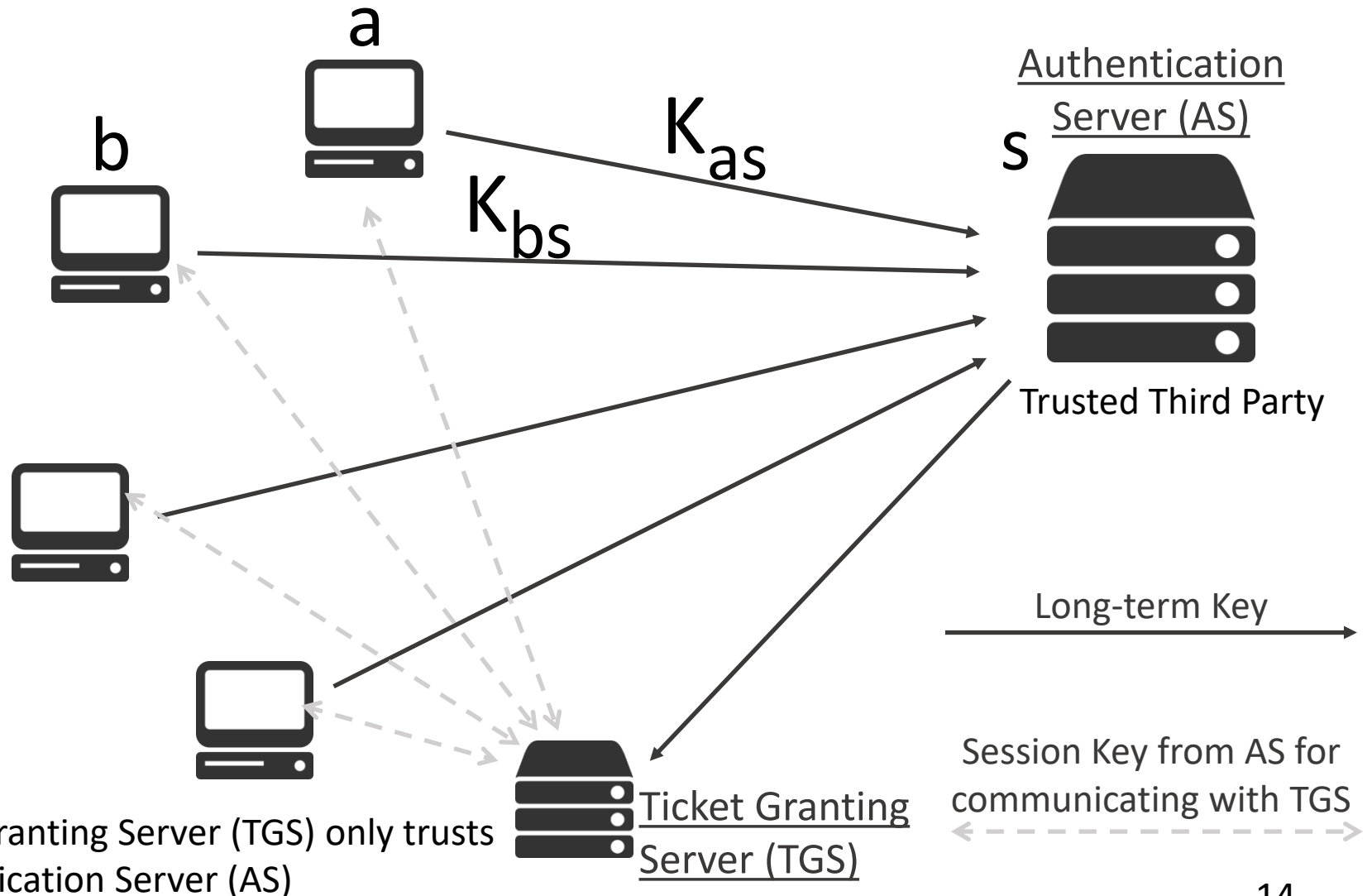


A new node is to be added to the network.

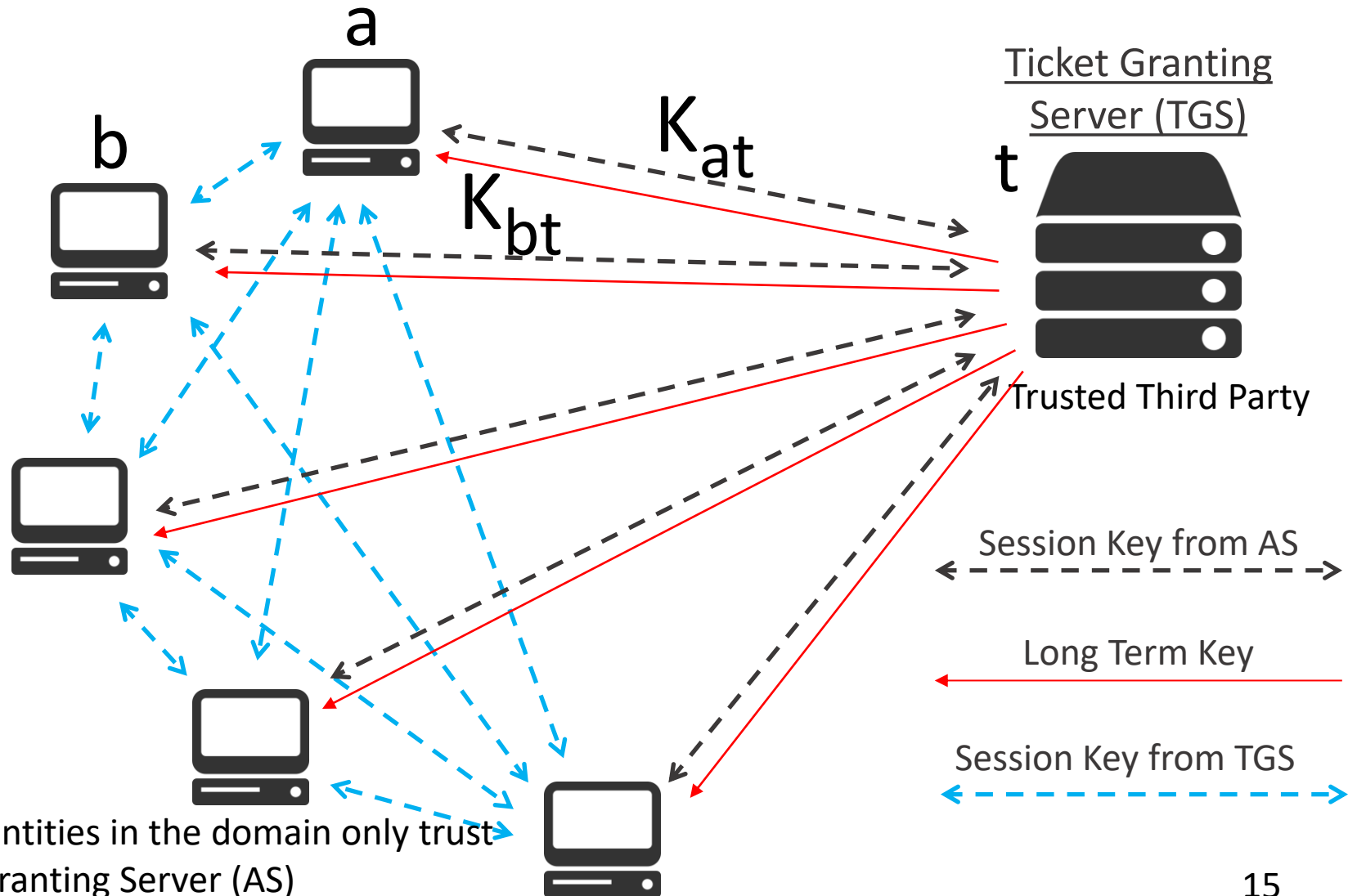
Since assigning a secret key to every pair of nodes is not practical, how should the nodes authenticate each other in a large network?

Introducing the concept of 'Trusted Third Party'

Authentication Server

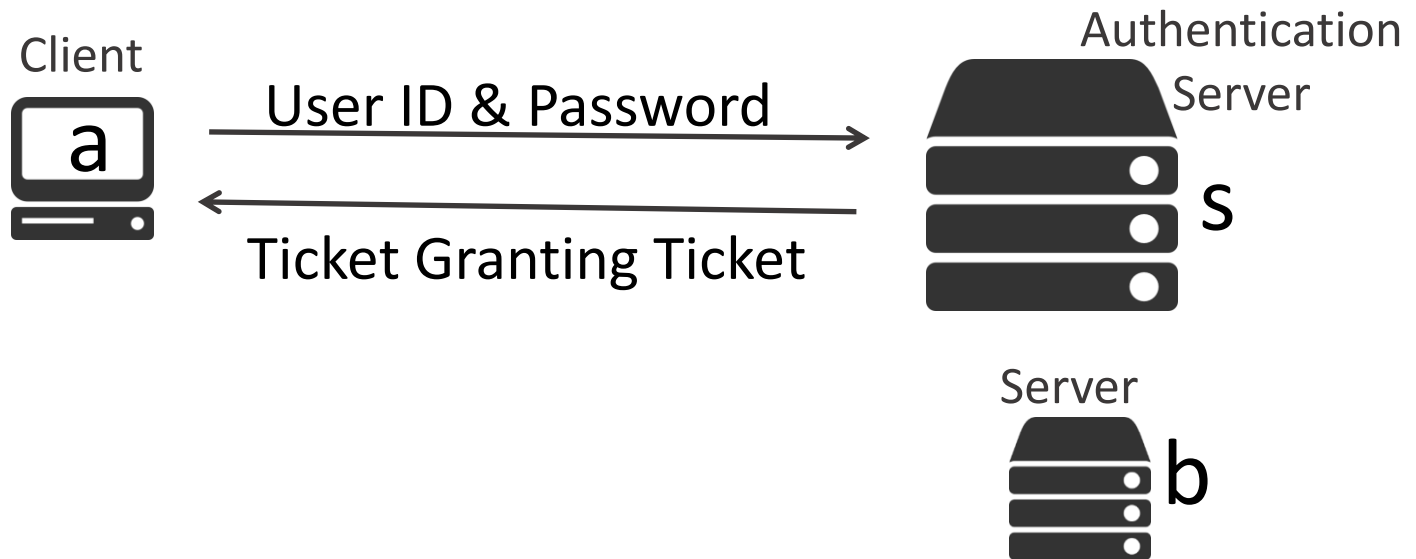


Ticket Granting Server



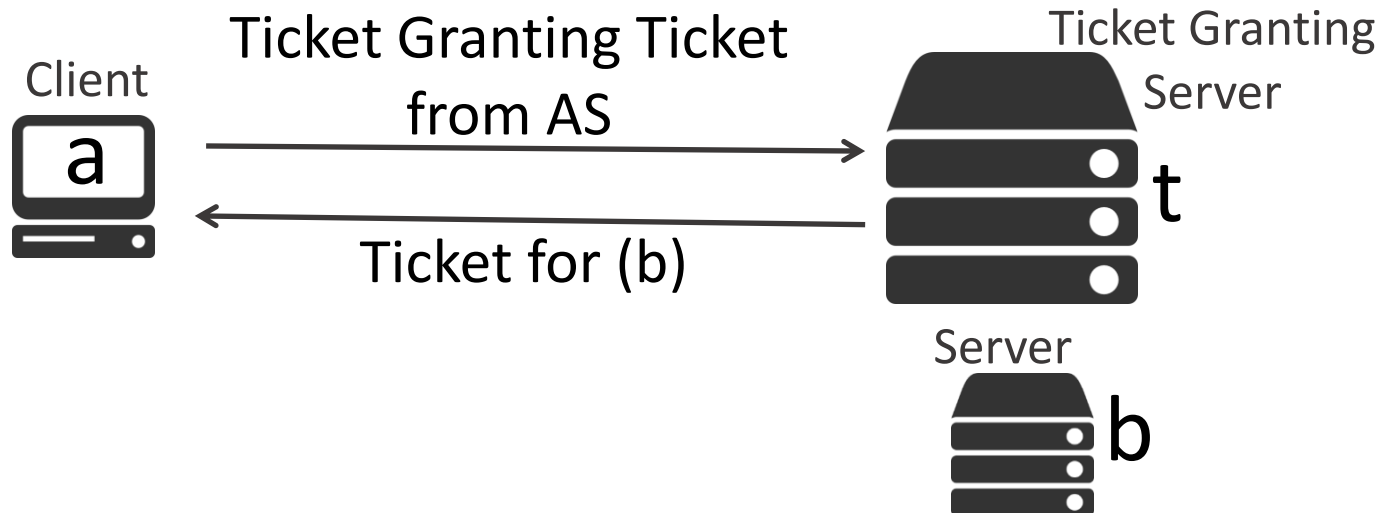
How does it work?

- Client (a) wants to get a service from server (b).
- Client (a) obtains a 'Ticket Granting Ticket' from the Authentication Server.



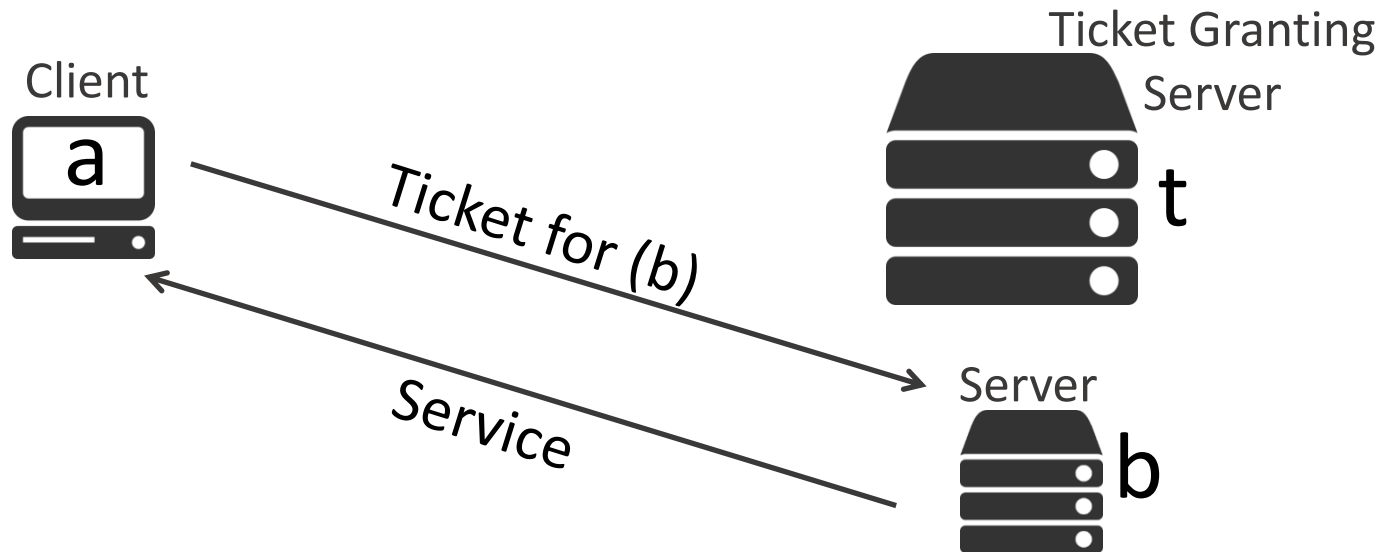
How does it work?

- Client (a) contacts Ticket Granting Server with the 'Ticket Granting Ticket' from the Authentication Server.
- Ticket Granting Server issues 'Ticket for (b)' to the client (a) stating it is approved to use the service provided by server (b).



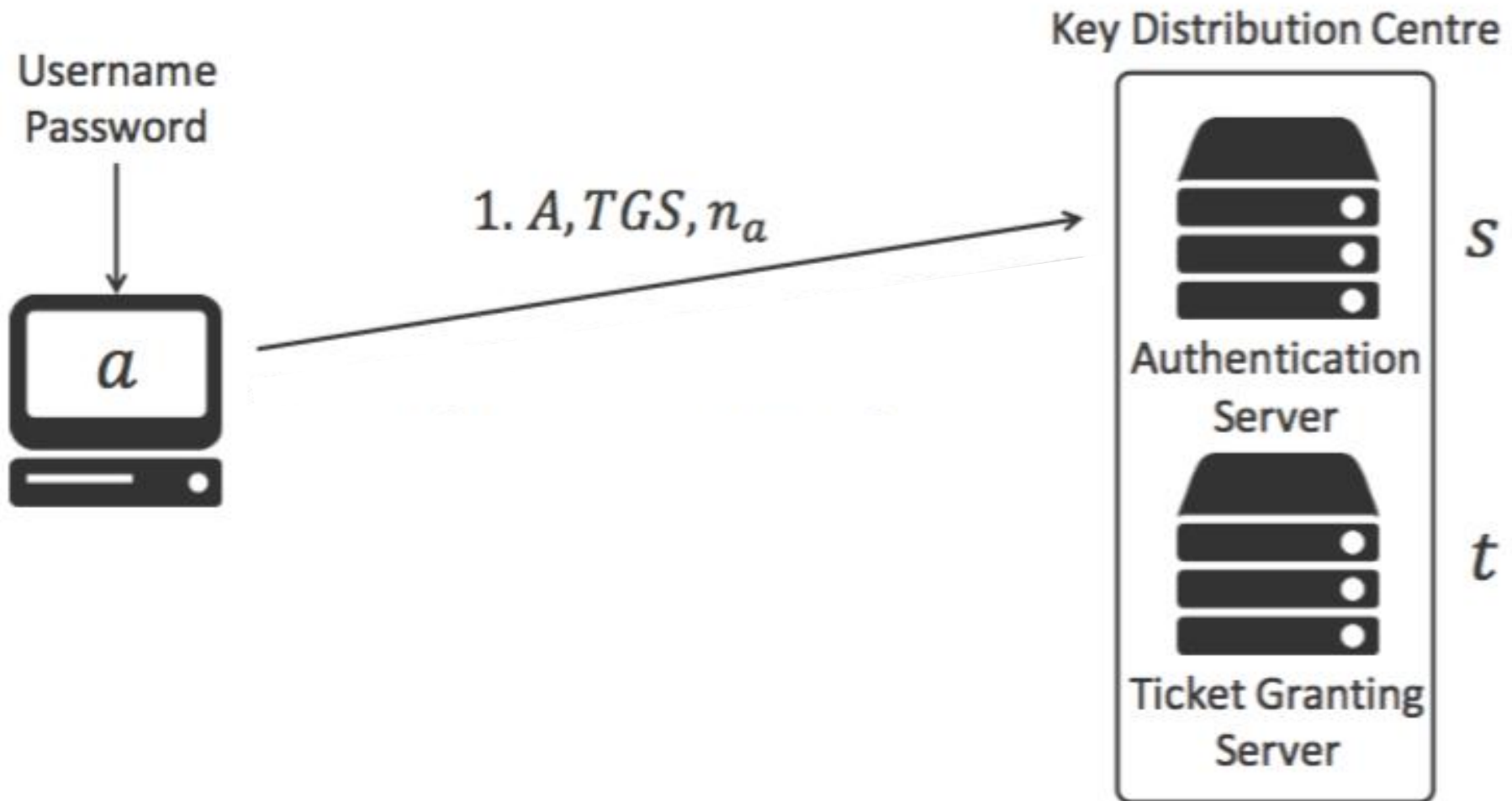
How does it work?

- Client (a) forwards the 'Ticket for (b)' to the server (b) requesting for a service.
- Client (b) allows client (a) to initiate the service provided by client (b) within a time frame.

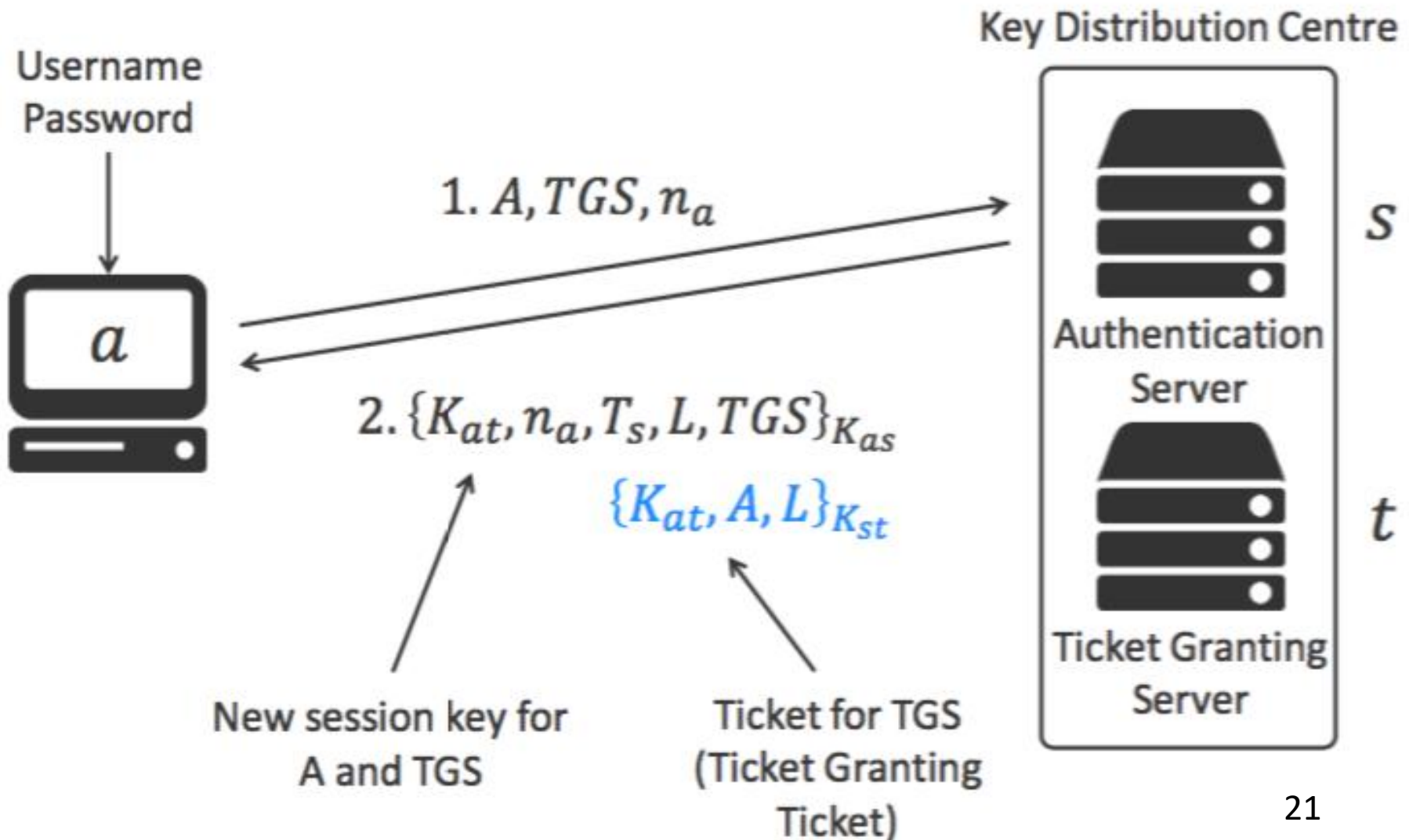


Let's put all the concepts together
for a detailed description about
Kerberos Authentication.

Kerberos: Step 1, Authentication

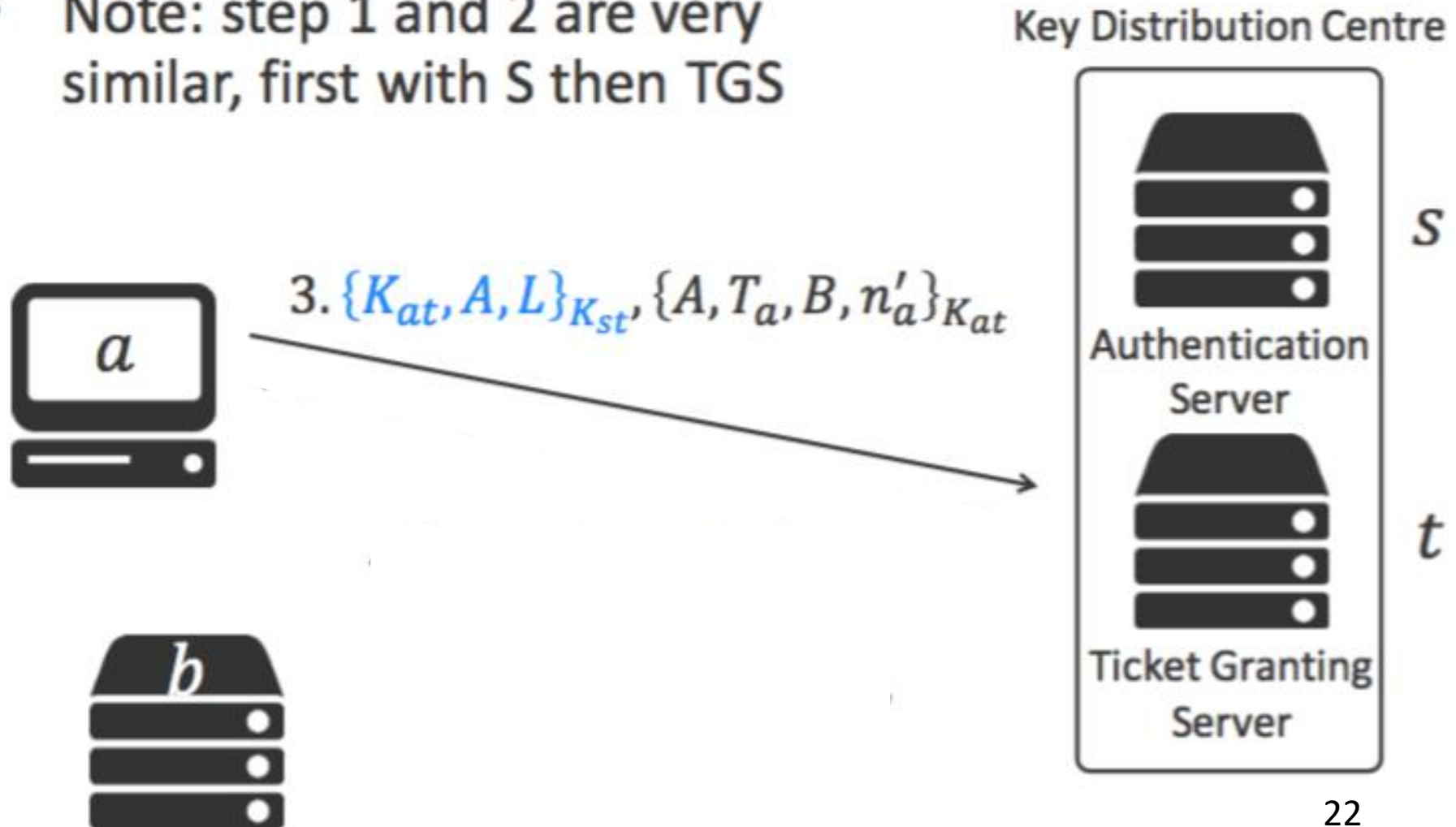


Kerberos: Step 1, Authentication



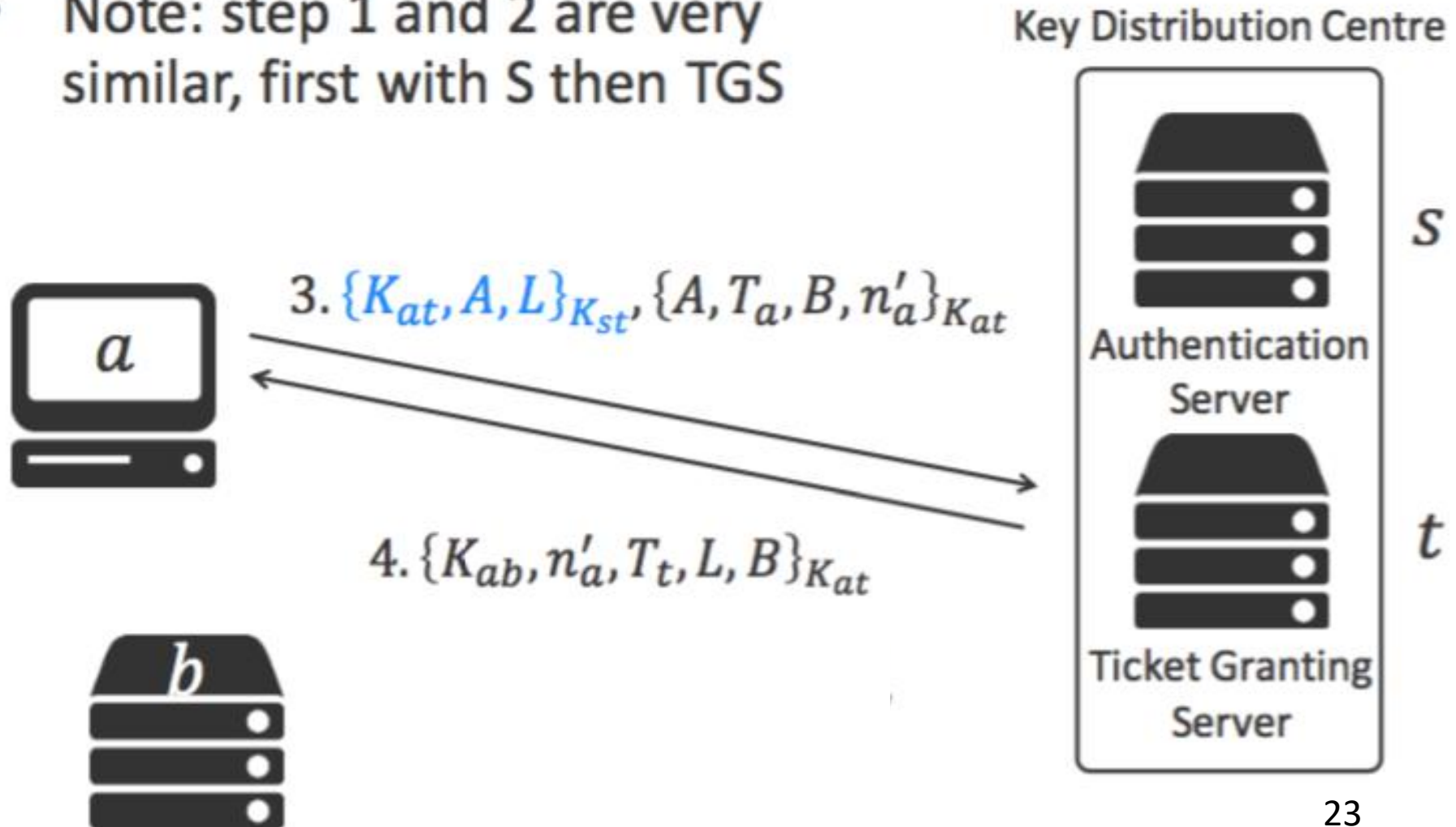
Step 2, Obtaining Tickets

- Note: step 1 and 2 are very similar, first with S then TGS



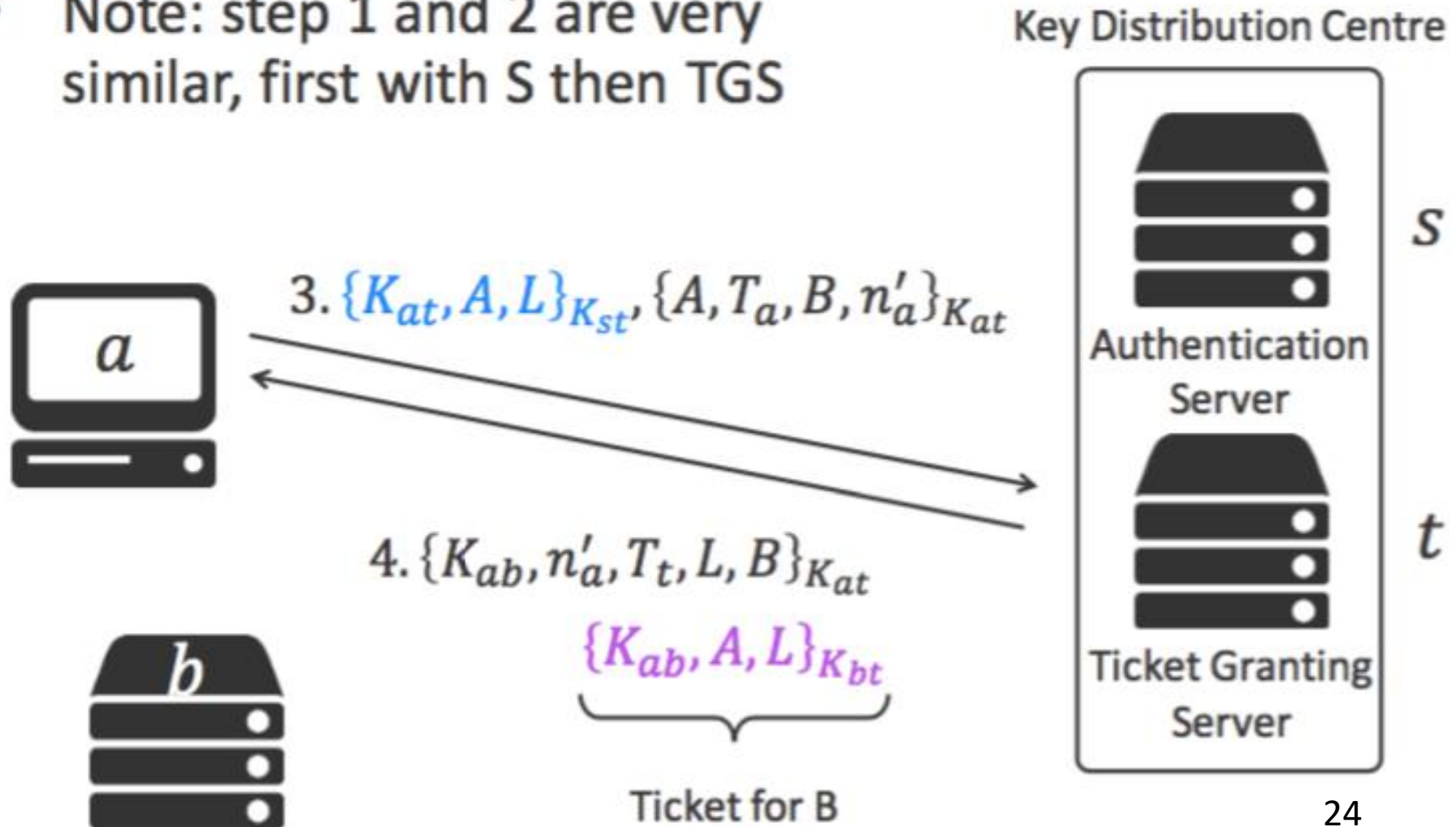
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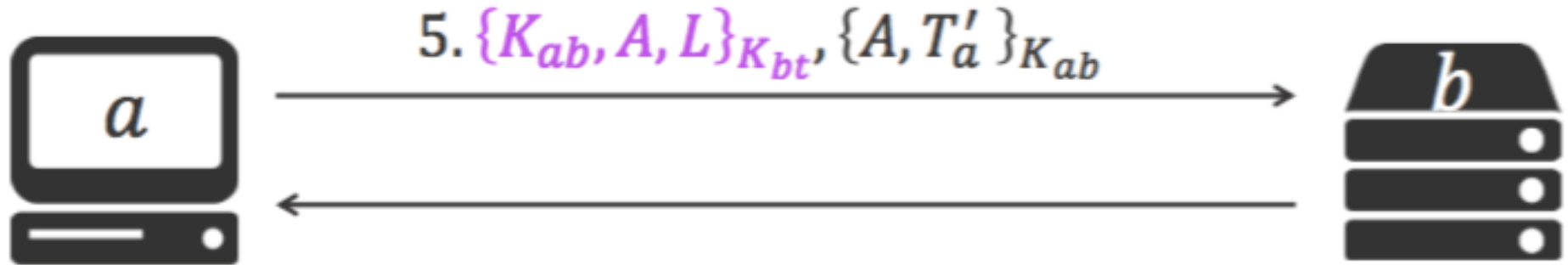


Step 2, Obtaining Tickets

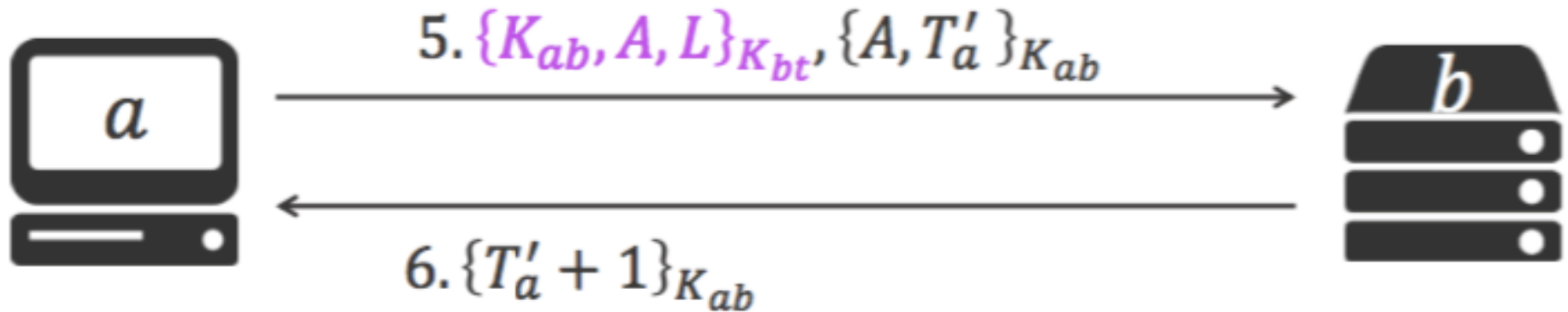
- Note: step 1 and 2 are very similar, first with S then TGS



Step 3, Using A Service



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Advantages of Kerberos

- Kerberos is open source.
- The availability of Kerberos on many recent operating systems such as:
 - Windows 2000 and above
 - Mac OS X
 - Red Hat Linux
 - Solaris

Drawbacks of Kerberos

- Kerberos requires the usage and the availability of a central server (Key Distribution Centre); if the server happens to go down then no one can login.
- To issue proper time-stamped tickets, hosts' clocks must synchronise properly in order for the protocol to work with the timed-stamped tickets used in the authentication process.

Conclusion

- To conclude, Kerberos is a network authentication protocol that allows proper authentication between a server and a client.
- It's a free and open source.
- It is an essential tool to provide optimum security while communicating between a server and a client.